

Diogo Sá

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Lisbon, Portugal

Work History

Game Programmer, ONTOP Studios

03/2023 – 09/2025

- Shipped iOS AR games including *New Fantasy* and *Devil's Purge*, progressing from contributing alongside a lead programmer to becoming the studio's sole programmer responsible for gameplay systems and technical implementation across new projects within a 4–5 person multidisciplinary team.
- Built gameplay systems from scratch including roguelike progression, inventory, encounters, dialogue, UI, geolocation, and player abilities using Unity and C# forming the core architecture of titles such as *Devil's Purge*.
- Created designer-facing tools using ScriptableObjects, Odin Inspector, Unity Visual Scripting, and state machines, enabling designers to author gameplay content independently.
- Integrated Unity with Node.js and MongoDB through REST APIs, implementing persistent player progression, quests, and in-game economy, including Unity IAP purchase validation and synchronization.
- Investigated runtime crashes, memory issues, and performance bottlenecks on iOS, using Unity Profiler and Xcode Instruments, improving mobile stability through Addressables asset loading, object pooling, and UI/audio optimization.
- Developed Pokémon GO-style geolocation gameplay using Online Maps asset and Mapbox, implementing point-of-interest interactions, route navigation prototypes, and location-based mini-games.
- Integrated a proprietary monetization and analytics SDK for *Ghostbusters: Afterlife ScARe*, implementing rewarded ads, interstitials, analytics events, and the required gameplay and UI integration for a live production update on Android.
- Developed custom HLSL shaders and extended Amplify Shader Editor with project-specific nodes to improve gameplay readability and support custom rendering effects.

Frontend Developer, Bliss Applications

07/2019 – 05/2021

- Summer internship followed by a part-time role during my MSc. Developed and maintained front-end features for client projects, collaborating closely with designers,

backend developers, and clients to deliver responsive and maintainable web applications.

- Developed and maintained responsive user interfaces using Vue.js, JavaScript, HTML, and CSS, implementing new features and maintaining existing production websites across multiple client projects.
- Integrated front-end applications with REST APIs, implementing data fetching, pagination, and asynchronous content loading to efficiently display media-rich content while maintaining responsive page performance.
- Maintained and enhanced existing web platforms using jQuery, JavaScript, and CSS, implementing UI improvements, bug fixes, and visual redesigns while working within established production codebases.
- Customized and extended the Zendesk support platform using HTML, CSS, and JavaScript, adapting the interface to match client branding and improve the customer support experience.
- Contributed to projects using modern front-end frameworks including Vue.js, React, and Angular, quickly adapting to existing codebases and development workflows across different client requirements.

Education

- Msc in Computer Science, Instituto Superior Técnico (Coursework completed) 09/2019 – 09/2023
- Bsc in Computer Science, Instituto Superior Técnico 07/2019 – 05/2021

Skills

- **Programming:** C#, C++, Python, JavaScript
- **Game / Graphics:** Unity, AR Foundation, OpenGL, HLSL, GLSL, Compute Shaders,
- **Backend:** Node.js, MongoDB
- **Web development:** HTML, CSS, Vue.js, Three.js, WebGL
- **Languages:** Portuguese (native), English (fluent)